GPS satellite relativistic corrections — standard derivation, proper-time companion, and open questions for a curved-spacetime extension

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§1 Cover note

This is a short follow-up to the interim report of 2026-05-25, narrowly scoped to one applied-relativity question that opened in the public repository the same week: the relativistic clock-correction problem for GPS satellites, prompted by a popular-science article on the headline $\pm 38\,\mu$s/day offset (issue #57). Two campaigns followed — a standard-method derivation (PR #71) reproducing Ashby's *Living Reviews* treatment end-to-end, and a proper-time companion (PR #73) that applies the Gill–Zachary substitution rules to each of the nine effects. The companion campaign surfaced four open questions about how the framework should extend to curved spacetime — questions the framework as-published does not yet answer. This report consolidates the two campaigns and the four questions in a form short enough to read alongside the broader interim report.

What this report is and is not

This is a *consolidation* of two campaigns of which the underlying work is committed to the public repository in full. It is not a re-derivation, not a journal-style summary, and not a request that you read the per-effect documents to interpret the questions in §5 — each question is self-contained. The report's load-bearing section is §5; everything else is context to make §5 readable in one sitting.

§2 The GPS question

GPS satellites carry atomic clocks that, once in orbit, tick faster than ground clocks by $+38.57\,\mu$s/day from the combined gravitational redshift (+45.65) and special-relativistic velocity dilation (−7.11). The engineering response is to slow the satellite clock by exactly this fractional rate before launch — the fundamental clock frequency of 10.230 000 000 00 MHz is built at 10.229 999 995 43 MHz, a fractional offset $\Delta f/f = -4.4647 \times 10^{-10}$. Once in orbit, the relativistic gain exactly cancels the pre-launch slowdown, and the satellite clock matches a ground clock at the geoid potential. This is the most stringently-tested applied-relativity result available: a deployed satellite constellation operating since 1977, with the relativistic correction validated daily against the $\sim 10^{-10}$ precision of cesium and rubidium clocks. Ashby (2003) is the canonical reference; the IAU 2000 geoid potential supplies the ground-side reference and the operational pre-launch offset is in the GPS Interface Control Document.

The proper-time framework's perspective is operationally aligned with how GPS actually works:

the satellite's atomic clock *is* its proper time τ, and the question "how does τ_{sat} relate to τ_{grnd}" is exactly the kinematic question the framework's $b = \sqrt{c^2 + \mathbf{u}^2}$ and $dt/d\tau = b/c$ apparatus is designed to answer. The campaign set out to determine, per-effect, whether the framework reproduces standard SR + GR at GPS-relevant precision (the expected outcome, given the framework's claim of mathematical equivalence), and to surface any structural difference that would be operationally distinguishable at the next-generation optical-clock precision floor ($\sim 10^{-17}$) or in regimes the framework as-published does not cover.

§3 Standard derivation — what PR #71 reproduces

The standard campaign covered nine separately-named effects, each in its own per-effect document with a Wolfram-MCP numerical check and a comparison against Ashby (2003). The headline numerical decomposition reproduces Ashby end-to-end at the precision the published parameters can sustain:

Effect	Standard-method magnitude	Ashby cross-reference
02 Gravitational time dilation	$+45.65\,\mu$s/day	§3.1 Eq. (19)
03 Velocity time dilation	$-7.11\,\mu$s/day	§3.2 Eq. (20)
04 Net headline + J_2	$+38.57\,\mu$s/day	§3.3 Eq. (39)
Pre-launch frequency offset	$\Delta f/f = -4.4647 \times 10^{-10}$	§3.3 Eq. (39)
05 Eccentricity periodic ($e = 0.02$)	± 46 ns peak	§4 Eq. (43)
06 Sagnac maximum	~ 137 ns per signal	§3.4 Eq. (35)
07 Shapiro maximum	~ 62 ps per signal	§6 Eq. (53)
08 J_2 contribution to geoid	$+0.033\,\mu$s/day	§3.5 Eq. (24)

The campaign's value to the repository is *consolidation* — collecting Ashby's per-effect derivations into one place with Wolfram-MCP verification of every quoted number from named parameter values (IERS / WGS-84 / EGM2008 / GPS ICD-200). This is a reproduction-of-standard exercise; no novel claim is made. We flag the campaign here because §4's proper-time companion is conditional on the §3 baseline being right.

§4 Proper-time companion — what PR #73 found

The companion campaign worked the same nine effects under the Gill–Zachary substitution rules ($\mathbf{w}/c = \mathbf{u}/b$, $(1/c)\partial_t = (1/b)\partial_\tau$, $b^2 = c^2 + \mathbf{u}^2$), recording per-effect whether the framework applies as-published or requires a speculative extension. The 9 per-effect documents split into three categories:

[OK] Direct framework application (pt_03, pt_06). Two effects where the verified substitution rules apply without modification. **pt_03 velocity time dilation** is the cleanest demonstration in the campaign: the framework's $d\tau/dt = c/b$ equals the standard $1/\gamma = \sqrt{1 - v^2/c^2}$ *exactly*, by algebraic identity from the velocity-duality substitution $\mathbf{w}/c = \mathbf{u}/b$. The numerical value $-7.11\,\mu$s/day is identical to standard; the agreement is at all decimal places, not at any precision floor. **pt_06 Sagnac** runs through the rotating-frame photon-null condition with $c \to b$ substitution; the result $\Delta t = -2\boldsymbol{\omega} \cdot \mathbf{A}/b^2$ reduces to the standard $-2\boldsymbol{\omega} \cdot \mathbf{A}/c^2$ at $b \approx c$ for an Earth-rotating receiver, with the residual at $\sim 10^{-10}$ ns (invisible to current GPS clocks, marginal at next-generation optical-clock precision).

[warn] Partial — kinematic [OK] + speculative GR extension (pt_01, pt_04, pt_05). Three effects where the kinematic piece is direct framework application but the gravitational piece requires a curved-spacetime extension that the framework as-published does not provide. The campaign adopts a *minimal-extension hypothesis* (replace c by the local b in the Schwarzschild line element) and shows it reduces to standard at GPS-relevant velocities. The combined headline $+38.57\,\mu$s/day is reproduced; the operational pre-launch frequency offset $\Delta f/f = -4.4647 \times 10^{-10}$ is unchanged. **pt_05 eccentricity** additionally surfaces a framework-specific *third-term* contribution $(\mathbf{u} \cdot \mathbf{a})/b^4 \cdot \tau_{\text{signal}}$ from the wave-equation dissipative coefficient in Maxwell-paper Eq. (4); for GPS circular orbits this vanishes ($\mathbf{u} \perp \mathbf{a}$), and for elliptical orbits at $e = 0.02$ it is $\sim 10^{-34}$ s per signal, operationally invisible. The same coefficient appears in the radiation-reaction sub-investigation #43 where it contributes at order unity — the framework is internally consistent on which structures appear where.

[X] Framework as-published doesn't extend (pt_02, pt_07, pt_08). Three effects that are purely gravitational (Schwarzschild redshift; Shapiro propagation delay; J_2 multipole). The framework's substitution rules govern SR Maxwell + flat-space dynamics; they do not extend to curved spacetime in any verified paper. Under the same minimal $c \to b$ Schwarzschild extension, each effect reduces to standard at GPS precision — but the speculative extension is the load-bearing assumption, not a derived result. Every doc in this category carries a substantive-AI TODO block on its §0 "Framework applicability" paragraph.

Operational summary. At GPS precision ($\mathbf{u}^2/c^2 \approx 1.7 \times 10^{-10}$; gravitational $GM/(rc^2) \approx 10^{-10}$; clock stability $\sim 10^{-14}$), the framework reproduces the standard prediction for every effect. The pre-launch frequency offset $\Delta f/f = -4.4647 \times 10^{-10}$ is unchanged; the operational engineering response is unchanged. **No observable deviation was found.** This is the *expected* outcome of mathematical equivalence in the appropriate limit, and is consistent with — but does not independently validate — either formulation. A *negative* finding (the framework predicting something different at GPS precision that disagreed with measurement) would have been a serious problem for the framework; none was found.

§5 Four open questions for a curved-spacetime extension

This is the load-bearing section of the report. The four questions all surface from the proper-time companion's [X] and [warn] per-effect documents; each is operationally invisible at current-generation GPS precision but structurally load-bearing for the framework's extension to regimes where the differences would matter. Each question is *one short paragraph* by design; the per-effect documents in `Roadmapping/GPS_Relativity/` carry the supporting context if you want to chase down details.

Q1. What is the correct curved-spacetime extension of the framework? The minimal-extension hypothesis used in pt_01, pt_02, pt_04, pt_07, and pt_08 is $c \to b$ in the Schwarzschild line element, with $b \to c$ at low velocity. This is *one* possible extension; we considered three others without resolution. Alternatives include (a) $c \to b$ applied only in the time-time metric component g_{00}, (b) treating null and timelike geodesics differently (the photon's local frame has $\mathbf{u} = 0$ and so $b = c$, but the worldline along which the metric is evaluated does not), and (c) replacing the metric structure entirely with a b-dependent connection that reduces to the Schwarzschild connection in the $b = c$ limit. For GPS at the 10^{-10} precision floor, all four hypotheses reduce to the same operational answer; the question matters for next-generation optical-clock GPS (10^{-17} stability is at the edge of distinguishing them) and for strong-field regimes where $GM/(rc^2)$ is no longer $\sim 10^{-10}$. The campaign cannot resolve this from the verified corpus and flags it as the principal Tepper-input ask of this report.

> **Q1 update — 2026-05-27, from issue #81.** A 5th candidate extension surfaced after this report's first draft: in the .tex source of your *Dual Newtonian Theory* paper (attached to issue #81), the framework's existing flat-space $V^2/(2mc^2)$ correction is applied to gravity via $V = -GMm/r$, producing the modified-Newtonian acceleration $\mathbf{a} = -(GM/r^2)(1 - GM/(c^2r))\hat{\mathbf{e}}_r$ (paper Eq. h4). This is structurally simpler than the four candidates above — no curved-spacetime apparatus, just the existing kernel applied to Newtonian gravity. **We worked the Mercury-perihelion test of this 5th candidate** (`Roadmapping/Mercury_Perihelion/` campaign): all algebra verifies [OK] via Wolfram MCP, and the numerical prediction is $\Delta\varphi_d = +37.79''$/century. **This under-predicts the observed $\approx 43''$/century by** 12%, and decomposes into a reduced-mass contribution $+44.66''$ (Corda's $\pi m/M$) plus a framework-relativistic correction of $-6.87''$ — opposite sign from GR's $+42.99''$. The 5th candidate is therefore ruled out by Mercury observation at the $\sim 10^{-1}$ level, narrowing the candidate space by one. The four extensions listed above remain open. Per-equation verification is recorded in `Equation_Verification/Dual_Newtonian_Theory.md`; the Mercury computation is in `Mercury_Perihelion/03_numerical_predictions.md`.

Q2. How does u·a transform under the proper-time boost (Maxwell-paper Eq. 11)? This is the same open question recorded in the `Roadmapping/Electromagnetism/_proper_time_cheatsheet.md` §4a (in the cheat-sheet's open-questions list, flagged for the radiation-reaction sub-investigation #43). It surfaces again in pt_05 of the GPS campaign: the third-term contribution $(\mathbf{u} \cdot \mathbf{a})/b^4$ to the eccentricity correction depends on $\mathbf{u} \cdot \mathbf{a}$ having a frame-independent operational meaning, and the covariance of this 3-vector dot product under the proper-time boost has not been derived in any verified paper. For GPS the question is moot ($\mathbf{u} \cdot \mathbf{a} = 0$ for circular orbits; $\sim 10^{-34}$ s for elliptical), but it bears on whether the radiation-reaction predictions of #43 are frame-independent at the level the experiments need. Even a one-line confirmation that $\mathbf{u} \cdot \mathbf{a}$ transforms as a Lorentz scalar (or, conversely, that it does not) would resolve this for both campaigns.

Q3. Does the framework predict a satellite-side Sagnac contribution? The standard Sagnac correction $\Delta t = -2\boldsymbol{\omega} \cdot \mathbf{A}/c^2$ carries c because it is derived in the rotating frame of the *receiver*. Under the proper-time framework, the receiver's effective light-speed is $b_{\text{recv}} = \sqrt{c^2 + u_{\text{recv}}^2}$ where $u_{\text{recv}} \leq 465$ m/s is the ground rotation, giving $b_{\text{recv}}^2/c^2 - 1 \approx 1.2 \times 10^{-12}$. The receiver-side derivation in pt_06 uses b_{recv} and recovers the standard prediction at this precision. A symmetric reading of the framework — in which the satellite's $b_{\text{sat}} = \sqrt{c^2 + u_{\text{sat}}^2}$ (with $u_{\text{sat}} \approx 3874$ m/s, giving $b_{\text{sat}}^2/c^2 - 1 \approx 1.7 \times 10^{-10}$) also enters the Sagnac formula on the source-side — would predict an additional satellite-side correction at the $\sim 10^{-10}$ of the standard Sagnac level. This is below current GPS precision but at the edge of next-generation optical-clock receivers. We ask whether the framework intends the source-side and receiver-side b to enter symmetrically, or whether the receiver-frame derivation is the complete prediction.

Q4. Are there framework-specific J_2-like corrections beyond the speculative extension? Earth's quadrupole moment $J_2 = 1.083 \times 10^{-3}$ modifies the gravitational potential via a $P_2(\cos\theta)$ multipole term. Under the speculative $c \to b$ extension, the J_2 correction enters via the same $1/b^2$ factor as the leading point-mass term, reproducing the standard $+0.033\,\mu$s/day contribution to the headline. A first-principles derivation of multipole-induced clock corrections from the framework's underlying dynamics would clarify whether the standard contribution is the leading-order term or whether additional b-dependent contributions arise at higher multipole order. The Lense-Thirring (frame-dragging) contribution from Earth's rotation — at the $\sim 10^{-16}$ level for current GPS but within reach of ACES on the ISS — sits in the same category: the framework's prediction would need a curved-spacetime gravitomagnetic extension that we have not derived. For GPS Q4 is sub-picosecond per day; for next-generation Earth-orbiting clock missions it is the leading post-Schwarzschild correction.

§6 Suggested next steps

Ordered by what would unblock the most downstream work:

1. **Resolve Q1.** Of the four questions, Q1 (the curved-spacetime extension) is the one that determines whether the other three have well-defined answers. Q3 and Q4 are partially dependent on it; Q2 is independent and propagates to issue #43 regardless. Even a directional answer — "the right extension is $c \to b$ everywhere", or "it's the $c \to b$ only in g_{00} variant", or "the framework's intended GR extension is not yet written down, but the right answer is the cleanest minimal coupling consistent with the Maxwell paper's substitution rules" — would let the GPS campaign convert the four [X] and [warn] per-effect documents from speculative to derived.

2. **Address Q2 across two campaigns.** A $\mathbf{u} \cdot \mathbf{a}$ transformation rule would close out the third-term contribution in both pt_05 (GPS) and the radiation-reaction sub-investigation #43. Since #43 is the regime where $\mathbf{u} \cdot \mathbf{a}$ is operationally observable, the GPS context is incidental but the cross-campaign consistency is the load-bearing reason to write the rule down.

3. **Defer Q3 and Q4 to next-generation precision.** Neither question affects current GPS operations. They are flagged here for completeness and for future-mission scope (optical-clock GPS, ACES, mission concepts at the 10^{-17}–10^{-18} clock-stability floor). If you signal interest in either, the campaign would open follow-on threads.

If you find this report useful in its form and would prefer a substantive theory thread on Q1 — a derived curved-spacetime extension of the framework, going through the dual-Dirac equation or the proper-time photon propagator — that is the natural next campaign, and we would scope it as a follow-on issue keyed off your direction. If the as-published framework's silence on GR is intentional (the framework is SR-and-flat-space by design, leaving GR to the standard GR apparatus), confirming that would let us close the four open questions in this report with that disposition recorded.

Thank you for the opening to send focused follow-ups in this form. The prior interim report's §6 r_e question remains the highest-load ask across all campaigns; this report's Q1 is the highest-load ask within the GPS scope but is a much smaller decision since GPS is operationally non-falsifying at current precision.